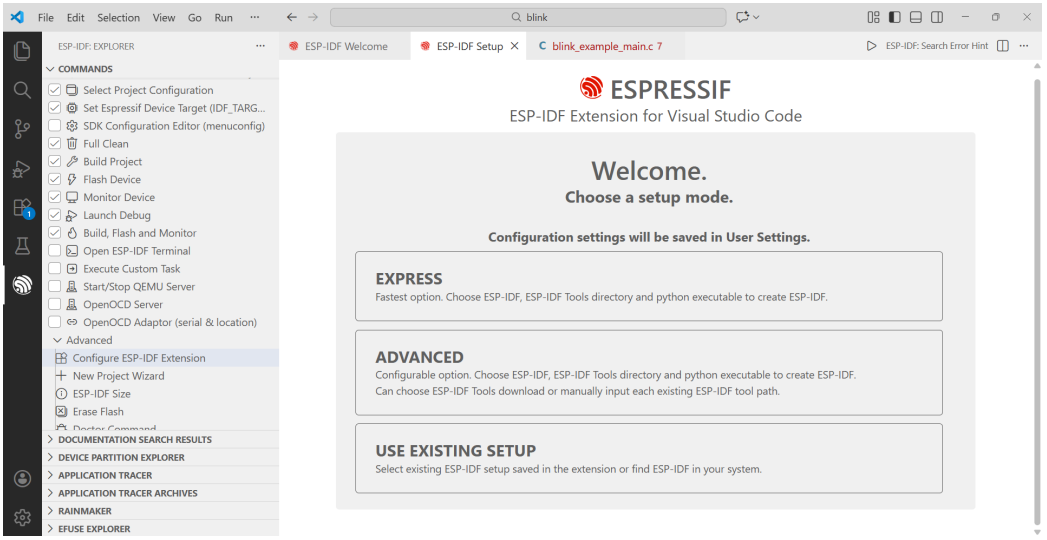
LED Blinking using Vaman board (ESP32)in

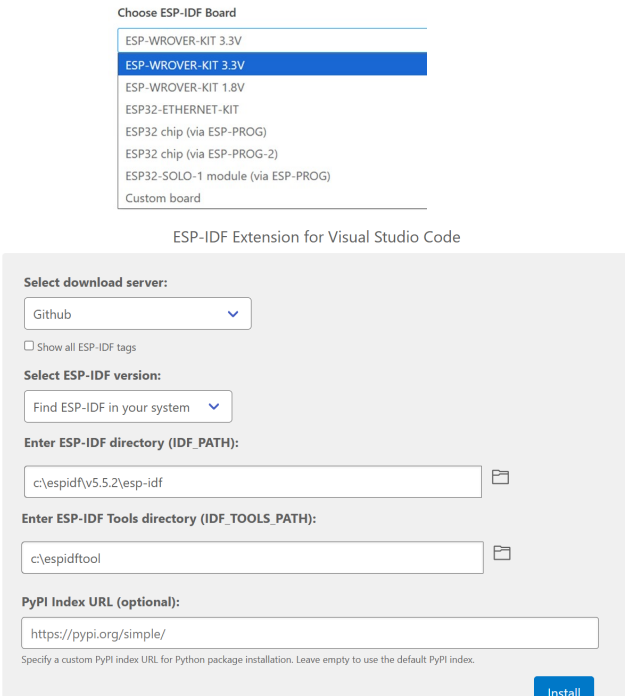
Windows in old VS Code version

- Install Python latest version from the link:
  Python: https://www.python.org/ftp/python/3.14.2/python-3.14.2-amd64.exe
- Install Git latest version from the link:
  Git:
  https://github.com/git-for-windows/git/releases/download/v2.52.0.windows.1/Git-2.52.0-64-bit.exe
- Install VS code:
  VS Code: https://code.visualstudio.com/download
- After installing the VS code

  - Click on **extensions** on the side task bar.
  - Search for esp-idf extension.
  - Install it and on the left task bar a new option of esp-idf installer appears in the last of the row.
  - Click on it > click on **Advanced** > click on **configure esp-idf extension.**
  - The below tab appears on the main screen.

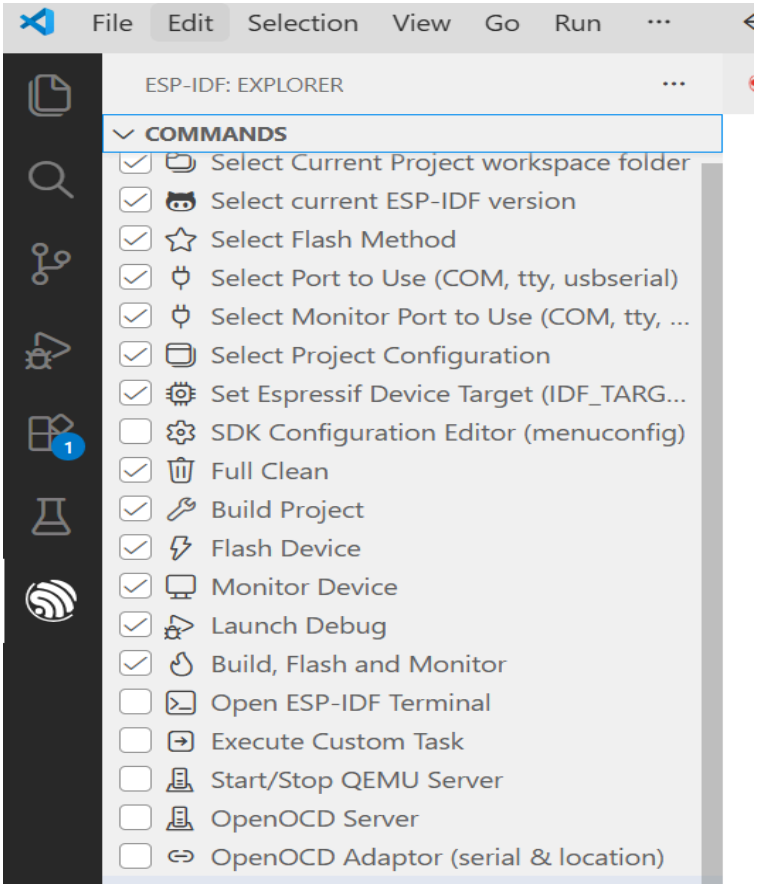


  - Click on Express> Select the ESP-IDF version (v5.5.2) > select the folder location where it should be saved. Select **esp32 chip via esp -prog -2**



- Click install.

- The installation will take approximately 20 minutes.
- After installing click on the file on the top task bar> **open folder**> click on examples> click **get-started**> select **blink** folder >**C:\espidf\v5.5.2\esp-idf\examples\get-started**.
  (select the folder where the esp-idf is saved).
- On the left task bar, in the commands section select the particular option and click on them to configure the required calibrations.



- Connect the vaman board with USB-Uart bridge from laptop to vaman board.

(1. If the com port is not getting detected then download the **CP210x_universal_windows_Driver** from google and unzip the downloaded file.

2. Now go to device manager and right click on the detected port **(other devices** dropdown**)** shown in the window, click on update driver > browse my computer> let me pick from a list>click on next > Click on **Have disk>** click on the previously unzipped file and select INF file**(silabser – setup information file).**

3. Click on ok and click on next

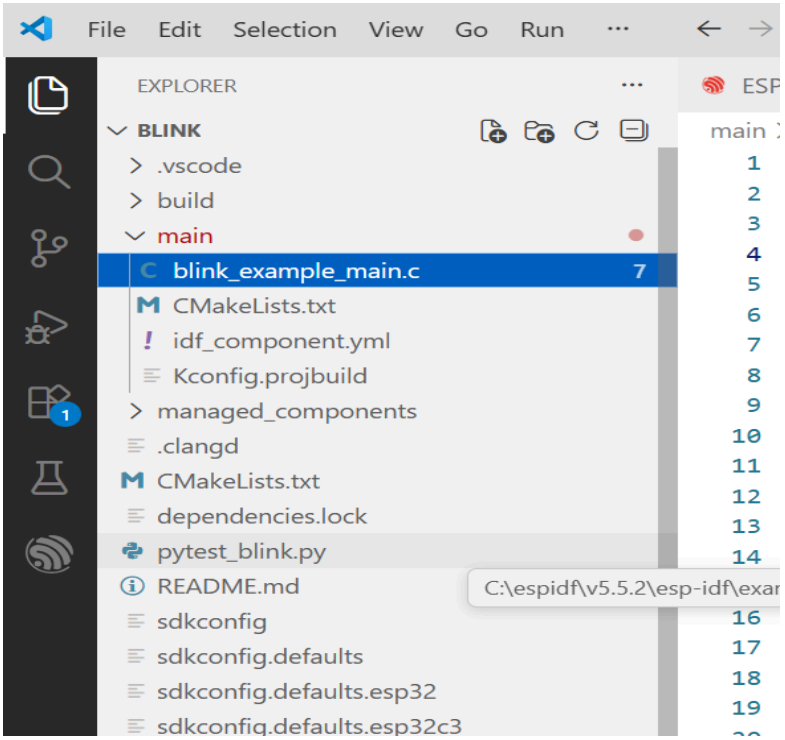
4. This will automatically enable the driver and now you can check the driver in device manager.

- Make sure the connections are proper. (Tx-Rx), (Rx-tx).

- Keep checking the bottom task bar for immediate updates about the connection. Also check with the device manager regarding the com ports while connecting the vaman board. Both the com port number in vs code and device manager should match. If you haven't found the option in the bottom task bar click on esp-idf extension and click on commands, a dropdown appears and click on the required settings such build project or flash method (if below bottom task bar wont appear).



- Click on the explorer option on the left side taskbar and click BLINK > main > select the .c file.



- Change the GPIO number which is matching for your board, **check the connection of the led properly and connect them.**

```
14  #include "led_strip.h"
15  #include "sdkconfig.h"
16
17  static const char *TAG = "example";
18
19  /* Use project configuration menu (
20     or you can edit the following li
21  */
22  #define BLINK_GPIO 26
23
24  static uint8_t s_led_state = 0;
25
26  #ifdef CONFIG_BLINK_LED_STRIP
```

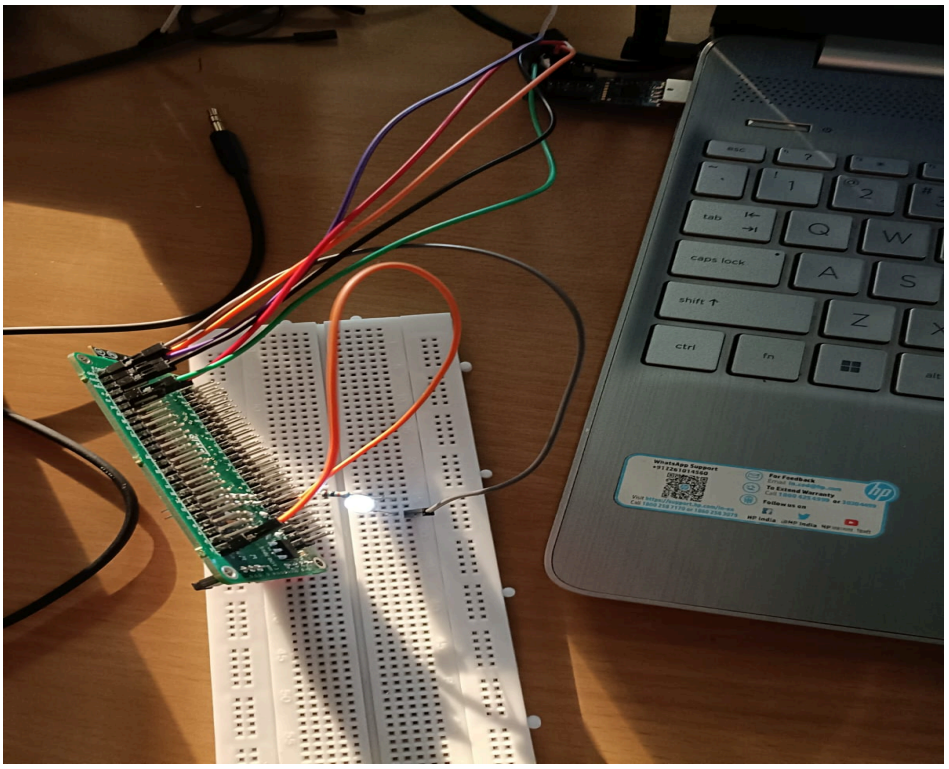
- Ctrl +s, save the changes.
- Click on **build** option on the bottom task bar.



- While clicking on the flash option get ready to unplug the GND – 0 jumper wire.
- Click on **flash** option and when the terminal gives a message such as percentage (10%, 30% etc...) ,then remove the wire.

```
Compressed 26304 bytes to 16559...
Writing at 0x00001000... (50 %)
Writing at 0x000075d8... (100 %)
Wrote 26304 bytes (16559 compressed
Hash of data verified.
Compressed 141760 bytes to 80281...
Writing at 0x00010000... (20 %)
Writing at 0x0001b35d... (40 %)
Writing at 0x00020f9e... (60 %)
Writing at 0x000267c4... (80 %)
Writing at 0x0002cd2d... (100 %)
```

- After removing the jumper wire and led starts flashing.
- Be care with the led connection.



Difficulty faced: While selecting the file select the file from **open folder** option, and not from **create project in vs code.**

MAC address:

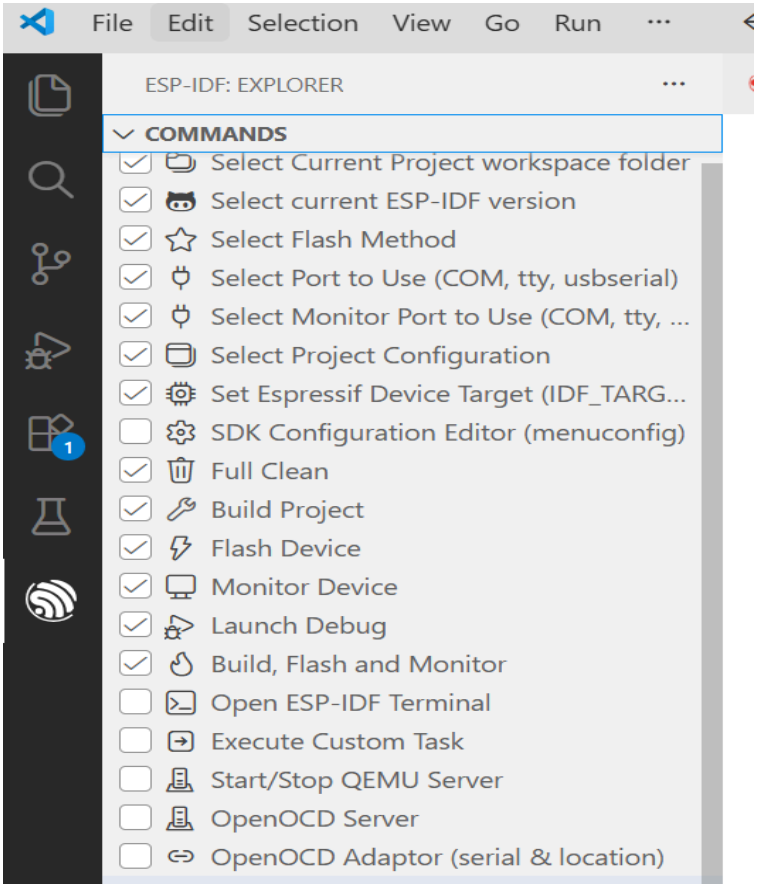
```
esptool.py v4.11.dev2
Serial port COM15
Connecting......
Chip is ESP32-D0WD-V3 (revision v3.0)
Features: WiFi, BT, Dual Core, 240MHz, VRef calibration in efuse, Coding Scheme None
Crystal is 40MHz
MAC: 54:43:b2:dc:72:74
Uploading stub...
Running stub...
Stub running...
Changing baud rate to 460800
Changed.
Configuring flash size...
Flash will be erased from 0x00001000 to 0x00007fff...
Flash will be erased from 0x00010000 to 0x00032fff
```

LED Blinking using Vaman board in

Windows in New VS Code version (Released on 04-02-2026)

- Install Python latest version from the link:
  Python: https://www.python.org/ftp/python/3.14.2/python-3.14.2-amd64.exe
- Install Git latest version from the link:
  Git:
  https://github.com/git-for-windows/git/releases/download/v2.52.0.windows.1/Git-2.52.0-64-bit.exe
- Install VS code:
  VS Code: https://code.visualstudio.com/download
- After installing the VS code
  - Click on **extensions** on the side task bar.
  - Search for esp-idf extension.
  - Install it and on the left task bar a new option of esp-idf installer appears in the last row.
  - Click on it and a new tab beside it opens.
  - Click on the **COMMANDS** option on the top of the page. A new dropdown appears.
  - Scroll down and click on the **ADVANCED** option on the new dropdown. This again opens a new drop down. Click on **Open ESP-IDF installation Manager.**
  - This opens a new window> click on **NEW INSTALLATION>** Click on **Start configuration Wizard.**
  - After navigating through the window, it asks for pre-requisites and requires downloading complete the following steps.
  - After completing Step 2 it opens step 3 with different target chips. Click on **esp32.**
  - In the next step for idf version installation select on **v5.5.2** and leave the default installation selections as it is.
  - Navigating through the next steps, continue with default settings for the entire process.
  - While in step 8 remember the destination of the saved folder. Complete the next installation with default settings.
  - This will automatically download all the requirements for vs-code esp-idf.
  - Now close all the tabs and reopen vs-code.

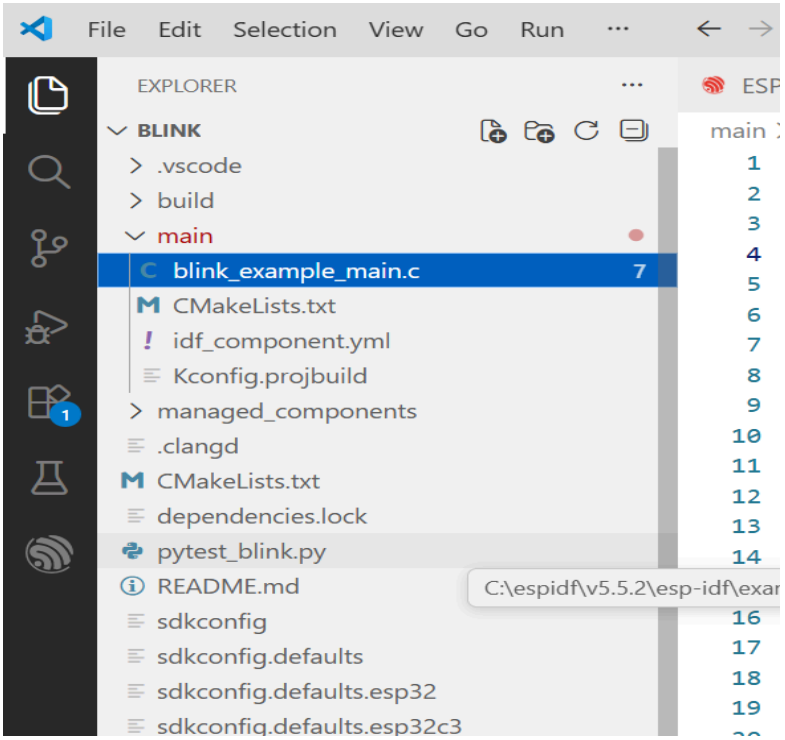
- The installation will take approximately 20 minutes.
- After installing click on the file on the top task bar> **open folder**> click on examples> click **get-started**> select **blink** folder >**C:\espidf\v5.5.2\esp-idf\examples\get-started (select the folder where the esp-idf is saved).**
- On the left task bar, in the commands section select the particular option and click on them to configure the required calibrations.



- Connect the vaman board with USB-Uart bridge from laptop to vaman board.
    - Make sure the connections are proper. (Tx-Rx), (Rx-tx).
- Keep checking the bottom task bar for immediate updates about the connection. Also check with the device manager regarding the com ports while connecting the vaman board. Both the com port number in vs code and device manager should match.



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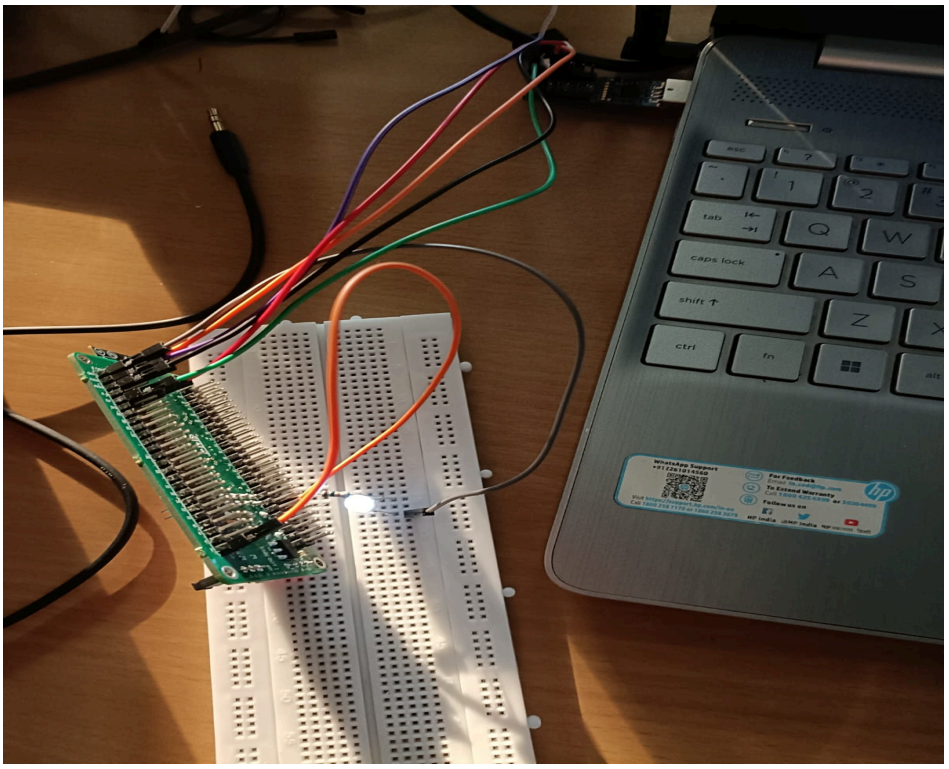
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